tiliqua: FPGA-based audio DSP platform

ORConf 2025

github.com/apfaudio/tiliqua

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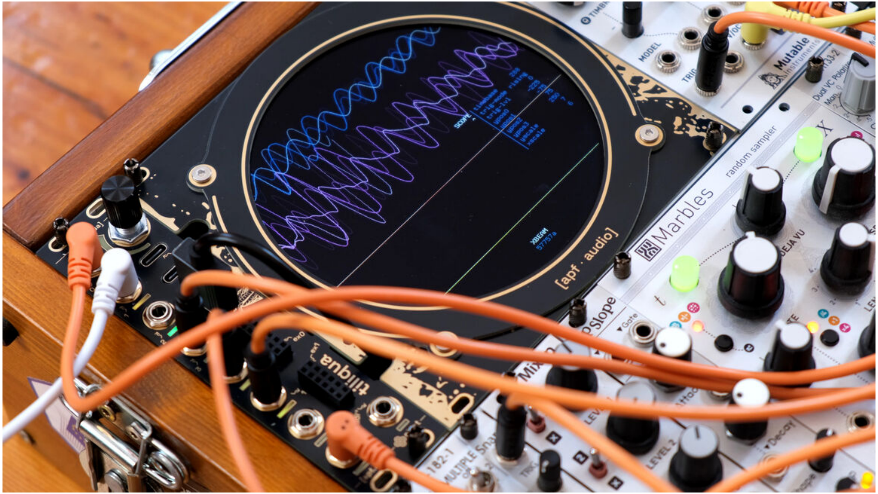
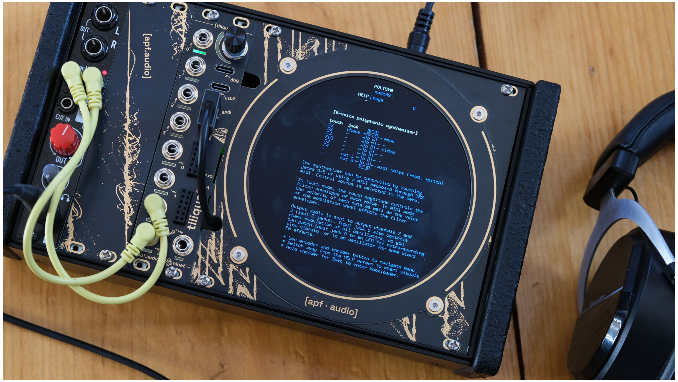
Eurorack?

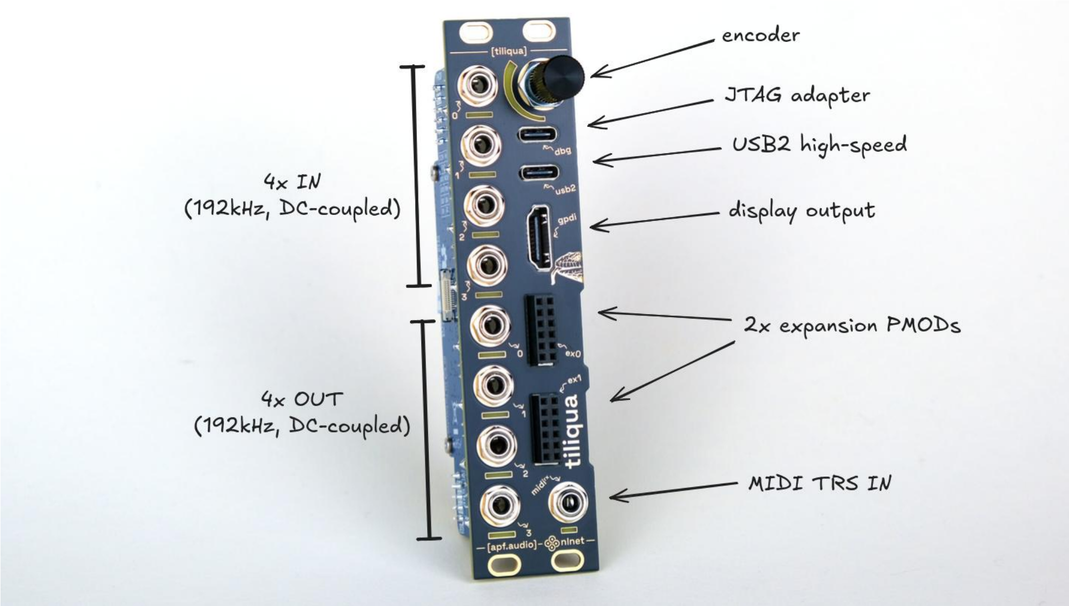


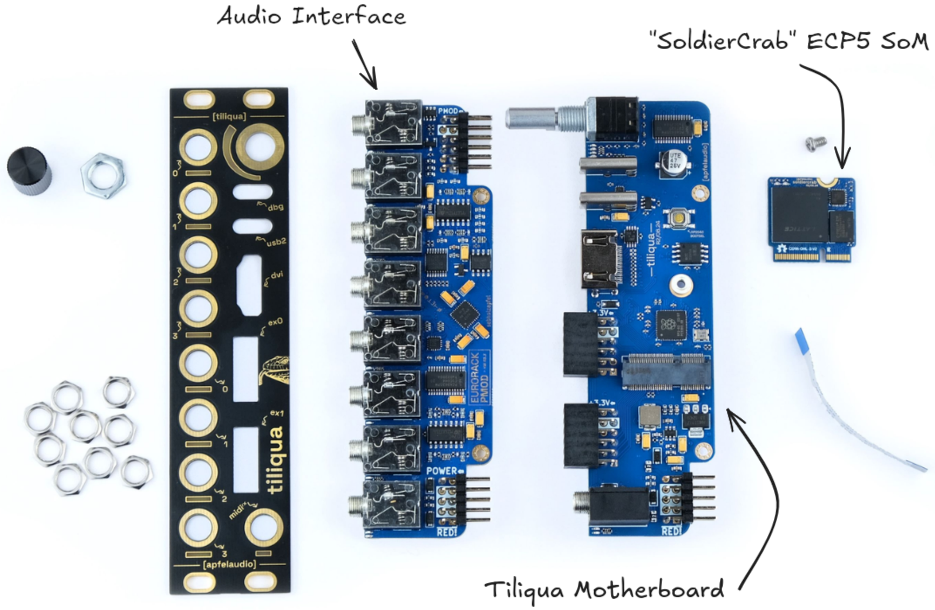
- Standard format for building synthesizers.
- **Eurorack ecosystem:** $> 15,000$ modules, $> 1,000$ different manufacturers.

What is tiliqua?



- **Open hardware platform for audio & video synthesis** (Eurorack module)
- **Open gateware library of DSP components** (Amaranth HDL)
- **Easy to share designs** – debugger built in, hot bitstream switching.
- **Useful to musicians, whether they touch HDL or not.**

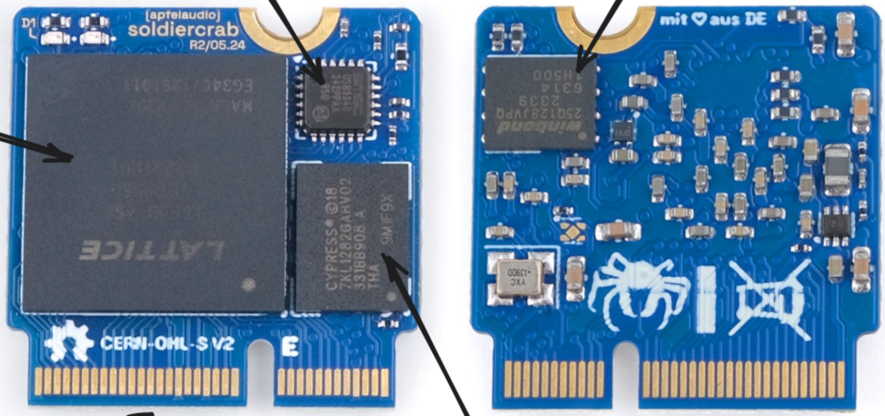

encoder
JTAG adapter
USB2 high-speed
display output
2x expansion PMODs
MIDI TRS IN
4x IN
(192kHz, DC-coupled)
4x OUT
(192kHz, DC-coupled)
[tiliqua]
dbg
usb2
gpdi
ex0
ex1
tiliqua
midi
[apf.audio]
nlnet


Audio Interface
"SoldierCrab" ECP5 SoM
Tiliqua Motherboard
[tiliqua]
tiliqua
[apfelaudio]
dbg
usb2
dvi
ex0
ex1
midi
PMOD
POWER
EURORACK PMOD
RED

USB2 HS PHY

128Mbit flash
(enough for 8 bitstreams)

Lattice ECP5
(LFE5U-25)



22mmx22mm

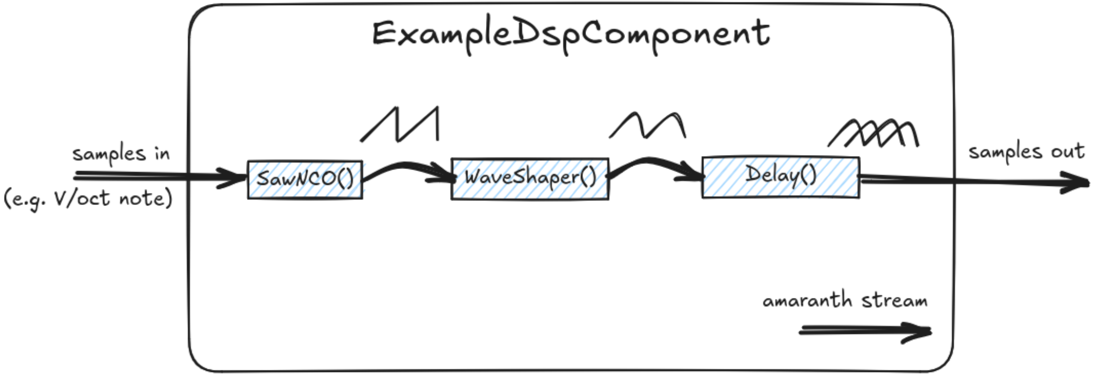
M.2 'MicroMod' compatible

256Mbit
HyperRAM / oSPI-RAM

`tiliqua` DSP library - some components

- **Resample** – Rational resampler.
- **PitchShift** – Granular pitch shifter.
- **DelayLine** – Memory-backed audio delay line.
- **Waveshaper** – Interpolating waveshaper.
- **FIR** – FIR filter.
- **VCA** – Voltage controlled amplifier.
- **NCO** – Numerically controlled oscillator.
- **Trigger** – Stream hysteresis.
- **Ramp** – Synchronized ramp generator.
- **SVF** – Tunable state-variable filter (HPF/BPF/LPF).
- **MatrixMix** – Matrix mixer with tweakable coefficients.
- **Boxcar** – Cheap averaging without multipliers.

DSP pipelines, Amaranth HDL and streams

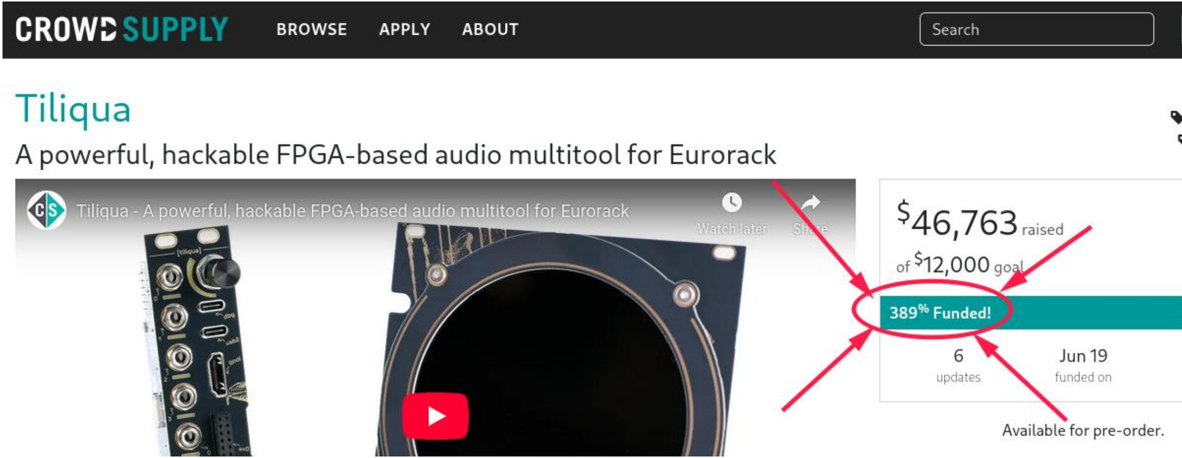


```python
class ExampleDspComponent(wiring.Component):
    i:  In(stream.Signature(fixed.SQ(1, 15)))
    o: Out(stream.Signature(fixed.SQ(1, 15)))
    def elaborate(self, platform):
        m = Module()
        # Create some DSP blocks ...
        nco    = dsp.SawNCO()
        shaper = dsp.WaveShaper(
                lut_function=sine_osc, lut_size=128, continuous=True),
        delay  = dsp.DiffusionDelay()
        # Connect them together ...
        connect(m, flipped(self.i), nco.i)
        connect(m, nco.o, shaper.i)
        connect(m, shaper.o, delay.i)
        connect(m, delay.o, flipped(self.o))
        m.submodules += [nco, ws, rs]
        return m
```

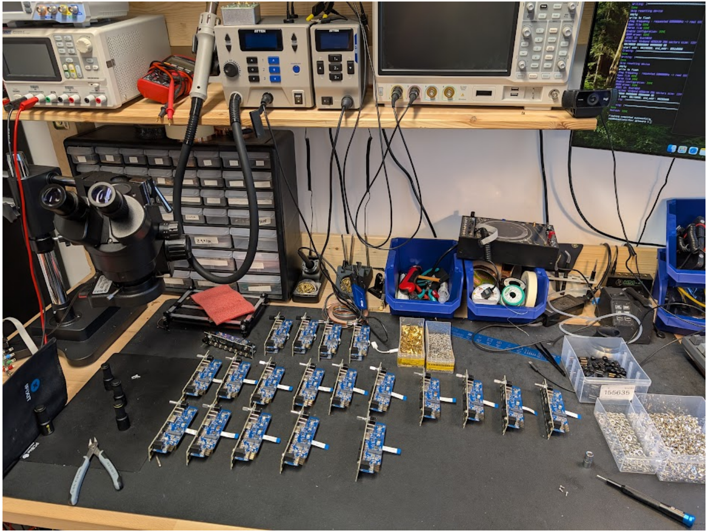
Past year… 3 big things

Past year... Funding!

Past year... Preorders!



Past year... Production!



Challenges...

- **More DSP cores, lib.fixed** – FFTs, STFTs, frequency-domain stuff!
- **Hardware/EMC/CE** – PCBA redesigns.
- **Dynamic pixel clocks** – hot-swap screens.
- **USB2 Host** – in pure gateware, based on LUNA.
- **Persistent options** – bitstream auto-restore on power loss.

`tiliqua` DSP library - frequency-domain processing (new!)

- **FFT** – forward + inverse, iterative, low area usage.
- **CORDIC** – complex/polar conversions.
- **Window** – windowing core and selectable window functions.
- **STFT** – forward + inverse, frequency-domain processing of time-domain signals.
- **SpectralEnvelope** – compute filtered power spectra over time.
- **SpectralCrossSynthesis** – vocoder-like effect.

Also: check out Amaranth `lib.fixed`

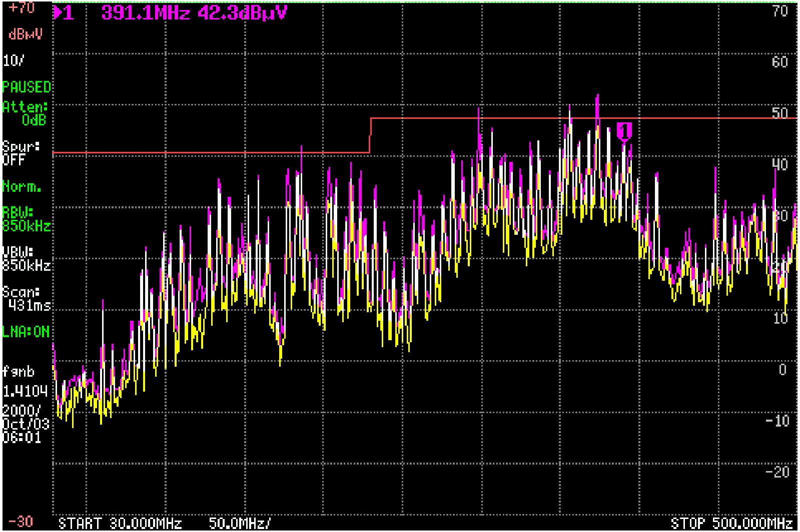
- **Extension to Amaranth standard library**
- **RFC + Implementation (WIP)**: github.com/amaranth-lang/amaranth/pull/1578
- **Comments welcome!**

- **FFT / `lib.fixed` Demo?**

Hardware / EMC: chamber

- Open source TEM Cell design (By Petteri Aimonen: **essentialscrap.com/tem_cell**)

Hardware / EMC: fail



Hardware / EMC: what worked

- Reroute Switchmode supplies
- Reduce FPGA pin drive strengths
- Reduce/remove split ground planes
- **Careful with long cables!**
- **Spread-spectrum PLL** (on ECP5!)

Dynamic clocking (new!)

- People have different screens (pixel clock).
- Same bitstream should work everywhere.
- Core reads display EDID, dynamically adapts resolution!
- Details: `https://github.com/apfaudio/tiliqua/pull/107`

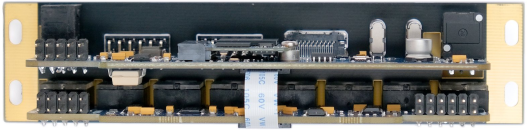
USB2 Host (new!)

- Cheapest MIDI keyboards only have USB MIDI :(
- Solution: pure-gateware USB2 high-speed host. (first?)
- Details: `https://github.com/apfaudio/tiliqua/pull/65`
- Builds on LUNA: `https://github.com/greatscottgadgets/luna`

Persistent Options (new!)

- Wear-levelled persistent storage to configuration flash
- Store last bitstream, all options, autoboot.
- Details: `https://github.com/apfaudio/tiliqua/pull/129`

Thank you for listening!



- contact: `seb@apf.audio`
- repository: `github.com/apfaudio/tiliqua`
- mailing list: `apf.audio`

(almost) shipping!

- `crowdsupply.com/apfaudio/tiliqua`